



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

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The algorithmic framework of this thesis builds directly on the Fisher-Orthogonal Projected Natural Gradient method of [Garg, Kolhe, Peng, and Gopalam \(2026\)](#), whose derivations and design served as the primary technical foundation for iFOPNG. The hypernetwork architecture, weight-chunking scheme, and functional regularizer are due to [Oswald, Henning, Grewe, and Sacramento \(2020\)](#) and are adopted here.

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Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse under hypernetwork architectures: the architectural necessity of weight-chunking forces the model to accumulate an exceptionally large gradient volume before a single projection step can execute, blowing up the condition number of the projection matrix and causing catastrophic inversion failures that standard regularisation cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce iFOPNG, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg et al., 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST, Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that iFOPNG consistently outperforms FOPNG across all benchmarks; in the standard-capacity hypernetwork setting only iFOPNG surpasses plain SGD, while most other projection variants fall below it. A standalone hypernetwork optimized with Adam and Oswald et al.’s functional

regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and `Weights & Biases`.

CODE. The baseline FOPNG framework was adapted from [Garg et al. \(2026\)](#); all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in $\text{\LaTeX}$ using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. <https://gemini.google.com>; Anthropic. (2026). *Claude* [Large language model]. <https://claude.ai>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* ([McCloskey & Cohen, 1989](#)). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Beyond its theoretical interest, this limitation carries direct practical cost. Deployed systems must either be retrained from scratch whenever the data distribution shifts — at substantial computational and energetic expense — or retain growing archives of historical data for replay, an option foreclosed in privacy-regulated and on-device settings where past data cannot be stored. Continual learning methods that preserve prior competence without replay therefore address both the economic burden of repeated retraining and the deployment of adaptive models under data-retention constraints.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2017) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by Oswald et al. (2020). Concurrently, within the field of geometric optimization, gradient projection methods, exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026), explicitly constrain subsequent updates to lie orthogonal to the subspace of parameter directions critical for previous tasks, with the geometry of orthogonality ranging from Euclidean (OGD) to Fisher-Riemannian (FOPNG).

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. Generating a target network’s full weight vector in one forward pass is computationally prohibitive, so chunked hypernetworks partition generation into K sequential sub-passes from a shared generator. Because every sub-pass executes before any gradient update, the gradients and Fisher estimates accumulated across chunks inflate with K , rendering the projection matrix numerically uninvertible. The exact algebraic mechanism is given in Section 4.5.1.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce *projection-scoped normalization* (column-wise ℓ_2 orthonormalization of G combined with absolute-maximum scaling of $\tilde{F}$) and iFOPNG, an inertial FOPNG variant using the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

What numerical pathologies arise when gradient projection methods are applied to chunked hypernetwork architectures, and do modifications to the Fisher metric construction—specifically an inertial combined metric and a monotonic accumulation operator—yield principled improvements over standard projection baselines in continual learning?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1 (Synergy vs. Conflict):** Does the joint integration of hard gradient projection with the hypernetwork’s soft functional regularizer yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?
- Sub-RQ2 (Gradient Pathology):** How does the architectural necessity of spawning K sequential weight-chunks inflate the gradient and Fisher magnitudes accumulated before a single projection step, and does rescaling both components independently within the projection algebra suffice to prevent matrix inversion collapse?
- Sub-RQ3 (Parameter Inertia):** Does replacing FOPNG’s task-specific metric F_{new} with the combined inertial metric $F_c = F_{\text{new}} + F_{\text{old}}$ yield consistent retention improvements across benchmarks?
- Sub-RQ4 (Fisher Accumulation):** Does accumulating F_{old} via a non-decaying element-wise maximum operator maintain long-horizon retention more effectively than an exponential moving average, and does the monotonic growth of the maximum operator impose a measurable plasticity cost over extended task sequences?

Summary of Empirical Findings

Projection-scoped normalization fully resolves the matrix inversion pathology, recovering stable projection performance across all seeds. The inertial combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ consistently improves retention over FOPNG, with the advantage scaling monotonically with inter-task Fisher overlap. The MAX accumulation operator outperforms EMA on long task sequences and requires no additional hyperparameters. In the standard-capacity hypernetwork setting, iFOPNG surpasses plain SGD but falls short of Adam; under extreme compression the gap to Adam widens to 20–40 pp and most projection methods drop below SGD entirely.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 Continual Learning and the Stability-Plasticity Dilemma

Modern machine learning systems are predominantly trained on fixed datasets drawn from a single stationary distribution — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (McCloskey & Cohen, 1989) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Task-incremental benchmarks are evaluated under one of two output-layer protocols. The *multi-head* protocol gives each task a dedicated output head, selected by task identity at test time, which removes output-level interference between tasks; the *single-head* protocol shares one output layer, forcing the backbone alone to resolve that interference. Farquhar and Gal (2018) showed that multi-head evaluation systematically inflates apparent method performance by eliminating a major source of forgetting, and argued that the single-head protocol is the more rigorous test. Split-MNIST is therefore evaluated under both: the multi-head variant reproduces the conditions of Garg et al. (2026) for direct numerical comparison, and the single-head variant follows Farquhar and Gal (2018) as the harder, more discriminating evaluation. The exact head configuration and the protocol assigned to each benchmark are given in Section 4.6.1.

3.3 Fisher Information Matrix

Several methods surveyed in this thesis quantify parameter importance through the *Fisher Information Matrix* (FIM). For a parametric model $p_\theta(y \mid x)$ with parameters $\theta \in \mathbb{R}^p$, the FIM is the $p \times p$ matrix

$$F(\theta) = \mathbb{E}_{x \sim q(x)} \mathbb{E}_{y \sim p_\theta(\cdot \mid x)} \left[\nabla_\theta \log p_\theta(y \mid x) \nabla_\theta \log p_\theta(y \mid x)^\top \right], \quad (1)$$

where $q(x)$ is the data-generating distribution (Amari, 1998). $F(\theta)$ defines a Riemannian metric on the parameter manifold: the Fisher inner product $\langle u, v \rangle_F = u^\top F(\theta) v$ measures distances in terms of the change in model predictions, not in terms of raw Euclidean parameter displacement. This metric is invariant under smooth reparameterization, making Fisher-norm constraints more principled than Euclidean-norm constraints for models whose parameters are not isotropically scaled.

The full FIM is intractable at the scale of neural networks. All methods evaluated in this thesis use the *diagonal empirical approximation*, which replaces the outer-product expectation with an empirical average over N observed data points:

$$\hat{F}_i \approx \frac{1}{N} \sum_{n=1}^N \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i} \right)^2, \quad (2)$$

yielding a non-negative scalar $\hat{F}_i$ for each parameter θ_i . The resulting vector $\hat{F} \in \mathbb{R}_{\geq 0}^p$ is treated as a diagonal matrix in all subsequent quadratic forms. This approximation reduces storage from $\mathcal{O}(p^2)$ to $\mathcal{O}(p)$ and is the variant used by EWC (Kirkpatrick et al., 2017) and all Fisher-based projection methods evaluated in (Garg et al., 2026).

3.4 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC uses F_i (Equation 2) as an importance weight for each parameter, and applies a soft quadratic penalty that resists changes to parameters deemed important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_{t-1,i}^*)^2, \quad (3)$$

where $\theta \in \mathbb{R}^p$ are the model parameters, θ_{t-1}^* is their snapshot saved after task $t-1$, and $\lambda > 0$ governs overall regularization strength. EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (De Lange et al., 2022).

3.5 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each parameter update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (4)$$

This provides a hard guarantee: the projected update cannot increase the loss on any previously seen task. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

However, standard Euclidean projection treats the parameter space as flat, assigning equal importance to all directions regardless of the curvature of the loss surface.

Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by $\hat{F}$, the diagonal empirical Fisher of the current task, so that the orthogonality constraint respects the local curvature of the loss

surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without any projection, and Orthogonal Natural Gradient (ONG) (Yadav, Mendoza, & Korrapati, 2025) combines Fisher preconditioning with Euclidean projection, sitting structurally between OGD and FOPNG. Exact algorithmic instantiations are given in Section 4.3.

3.6 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2017): a meta-model $\mathcal{H}(\cdot; \phi)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \phi)$ of a target network, conditioned on a task embedding e_t . Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task. Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (2020) introduced *weight chunking*: the generator iterates over C chunk embeddings $c_1, \dots, c_C$, producing a fixed-size block of target parameters per pass and concatenating them into the full weight vector θ_t . To prevent the hypernetwork from forgetting how to generate the weights of earlier tasks, a soft *functional regularizer* anchors the generator in its own output space. Let $\theta_s = \mathcal{H}(e_s; \phi)$ be the target-network weight vector the generator currently produces for task s under the single, evolving parameter set ϕ . On completion of task s , the generated vector is cached as a fixed reference θ_s^{ref} , detached from the computation graph. While task t is trained, only ϕ is updated, and every reference $\{\theta_s^{\text{ref}}\}_{s < t}$ stays fixed at the value it held when its task finished, under the penalty

$$\mathcal{L}_{\text{reg}} = \frac{\beta}{t-1} \sum_{s=1}^{t-1} \|\mathcal{H}(e_s; \phi) - \theta_s^{\text{ref}}\|_2^2, \quad (5)$$

with regularization strength $\beta > 0$. The penalty therefore measures, in the space of generated weights, how far the current generator has drifted from the exact target vectors it once produced for each past task. This is EWC applied at the meta level: rather than anchoring individual target parameters, it anchors the generated weight configurations for each previously seen task, penalising drift in the function space where task-relevant quantities actually live.

3.7 Identified Gaps and Motivating Pathologies

Whether hard geometric projection constraints are compatible with the soft functional regularization already employed by hypernetworks, or whether layering both simultaneously over-constrains the generator’s plasticity, has not been investigated (Sub-RQ₁).

The architectural necessity of weight-chunking introduces a gradient accumulation pathology — inflated condition numbers in the projection kernel — that is unique to the hypernetwork and projection methods combination and has no prior treatment in the literature (Sub-RQ₂).

Existing Fisher-based projection methods define the projection geometry from a single-task Fisher metric, which does not weight parameters by their importance to past tasks; whether replacing it with a combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ that anchors historically important coordinates, giving them inertia, yields consistent retention improvements across benchmarks has not been examined (Sub-RQ₃).

The historical Fisher estimate underlying Fisher-based projection depends on a choice of accumulation operator; whether a monotonic maximum or a decaying exponential moving average better preserves long-horizon retention has not been directly compared (Sub-RQ₄).

These four gaps jointly motivate the sub-research questions stated in Section 2 and the experimental programme described in Section 4.6.

4 METHODS

Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 Continual Learning Protocol

4.1.1 Task Setup and Evaluation Metrics

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$ presented in order. At task t , only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}, \quad (6)$$

and *backward transfer*

$$\text{BWT} = \frac{1}{T-1} \sum_{t=2}^T (a_{T,t} - a_{t,t}), \quad (7)$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i . Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

Reporting both metrics jointly is necessary because each can mask failure modes invisible to the other: a method that forecloses new learning to preserve old tasks achieves near-zero BWT alongside collapsed accuracy, while a method that overfits each new task sustains competitive accuracy at the cost of large negative BWT on earlier ones.

4.2 Fisher Information Matrix and Regularization-Based Continual Learning

All Fisher-based methods evaluated in this thesis use the diagonal empirical approximation F defined in Equation (2). The F_{new} and F_{old} used in projection methods denote the FIMs computed from the current task’s data and accumulated from all prior tasks, respectively; both are instances of Equation (2).

4.2.1 Elastic Weight Consolidation

EWC (Section 3.4, Equation 3) is included as a regularization baseline that operates purely through parameter-space penalties and maintains no gradient subspace. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 Gradient Projection Methods

This subsection presents the full suite of gradient-based methods evaluated in this thesis. SGD and FNG establish unconstrained lower bounds with no forgetting-protection mechanism; OGD through iFOPNG form the projection family that is the primary focus of this work.

4.3.1 Stochastic Gradient Descent

SGD applies vanilla gradient descent with no forgetting protection:

$$\Delta\theta_{\text{SGD}} = -\eta \nabla_{\theta} \mathcal{L}_t. \quad (8)$$

Adam (Kingma & Ba, 2015) is included as an adaptive-rate variant that maintains per-parameter momentum; both serve as unconstrained baselines.

4.3.2 Orthogonal Gradient Descent

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and updates parameters via the current gradient projected onto the orthogonal complement of $\text{Col}(G)$:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g, \quad \Delta\theta_{\text{OGD}} = -\eta \tilde{g}. \quad (9)$$

The projection is measured in Euclidean space, treating all parameter directions symmetrically regardless of their importance to the loss surface.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns; insertion, orthonormalisation, and overflow recompression are detailed in Section 4.5.1.

4.3.3 Natural Gradient and Fisher Natural Gradient

Natural gradient descent (Amari, 1998) replaces the Euclidean gradient step with one measured in the Riemannian geometry of the Fisher metric, so that equal step sizes correspond to equal changes in model predictions rather than equal changes in parameter values:

$$\Delta\theta_{\text{nat}} = -\eta F_{\text{new}}^{-1} \nabla_{\theta} \mathcal{L}. \quad (10)$$

FNG (Garg et al., 2026) applies a trust-region constraint to this step, normalising by the Fisher norm of the gradient:

$$v_{\text{FNG}}^* = -\eta \frac{F_{\text{new}}^{-1} g}{\sqrt{g^\top F_{\text{new}}^{-1} g}}. \quad (11)$$

FNG maintains no gradient subspace and provides no forgetting protection; it is included to isolate the contribution of Fisher preconditioning independently of projection.

4.3.4 Orthogonal Natural Gradient

ONG (Yadav et al., 2025) extends OGD by replacing the raw gradient with the natural gradient before projection. The natural gradient $F_{\text{new}}^{-1}g$ is first computed, then projected onto the Euclidean orthogonal complement of $\text{Col}(G)$:

$$\Delta\theta_{\text{ONG}} = -\eta \left(F_{\text{new}}^{-1}g - G(G^\top G)^{-1}G^\top F_{\text{new}}^{-1}g \right). \quad (12)$$

The step direction is Fisher-preconditioned, but the orthogonality constraint is still measured in Euclidean space: the kernel being inverted is $A = G^\top G + \lambda I$, identical to OGD. Following the diagonal Fisher approximation used throughout this thesis, we substitute the EKFA inverse of [Yadav et al. \(2025\)](#) with the diagonal empirical Fisher; results are not directly comparable to the original implementation.

4.3.5 Fisher-Orthogonal Projected Natural Gradient Descent

FOPNG ([Garg et al., 2026](#)) moves both the step and the projection into the Riemannian geometry of the Fisher metric. Where ONG projects the natural gradient in Euclidean space, FOPNG measures orthogonality in Fisher space: a direction is orthogonal to the stored basis if it produces no change in model predictions along those directions, not merely no change in parameter values. The constrained trust-region update is:

$$v_{\text{FOPNG}}^* = -\eta \frac{F_{\text{new}}^{-1} P g}{\sqrt{(P g)^\top F_{\text{new}}^{-1} P g}}, \quad P = I - F_{\text{old}} G \left(G^\top F_{\text{old}} F_{\text{new}}^{-1} F_{\text{old}} G + \lambda I \right)^{-1} G^\top F_{\text{old}}, \quad (13)$$

where F_{new} is the current-task empirical Fisher, F_{old} is the accumulated historical Fisher under which the stored directions were collected, and λI is a damping term for numerical stability. This is the pre-Fisher form of [Garg et al. \(2026\)](#), in which the stored basis carries its collection-time Fisher weighting.

4.3.6 Inertial FOPNG

In FOPNG the Riemannian metric governing both the trust-region constraint and the projection objective is F_{new} alone. This is natural on the first task but creates an asymmetry as history accumulates: the subspace $\text{Col}(G)$ encodes directions important to prior tasks, yet the metric assigns zero curvature to parameters that were important previously but are uninformative on the current task, allowing the update to move freely through historically sensitive directions.

iFOPNG resolves this asymmetry by replacing F_{new} with the *combined metric*

$$F_c := F_{\text{new}} + F_{\text{old}}, \quad (14)$$

where F_{old} is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Substituting F_c for F_{new} throughout Equation (13) yields:

$$v^* = -\eta \frac{F_c^{-1} P g}{\sqrt{(P g)^\top F_c^{-1} P g}}, \quad P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}, \quad (15)$$

where $A = G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G + \lambda I$. A formal derivation is provided in Appendix A.

The combined metric assigns each coordinate i an effective curvature $(F_{\text{new}})_i + (F_{\text{old}})_i$, suppressing the step size in directions important to any task in the sequence. We call this *parameter inertia*: a coordinate that accumulated high Fisher importance is not pulled back toward a stored reference point — it is made harder to displace. This contrasts with EWC (Kirkpatrick et al., 2017), which achieves retention through an explicit quadratic penalty anchored at θ^* ; iFOPNG achieves it geometrically, through the ambient metric alone.

4.4 Hypernetwork Architecture

4.4.1 Chunked Hypernetworks

Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k , $k = 1, \dots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 Functional Regularizer

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 Target Network and Dropout

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two Dropout($p=0.5$) layers in the SimpleCIFARCNN classifier head. Dropout is removed from the target network in all hypernetwork configurations. In chunked generation, the hypernetwork executes K sequential forward passes through the same target architecture before any gradient update; because all passes share the parent module’s training state, stochastic dropout masks are resampled independently on every chunk pass, injecting uncontrolled noise into the gradient signal accumulated across chunks. Removing dropout eliminates this source of variance without modifying any other aspect of the target architecture

4.5 Novel Algorithmic Contributions

4.5.1 Projection-Scoped Normalization

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K , inflating the condition number of the projection kernel $G^\top FG$ and causing the matrix inversion in Equation (13) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G :

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \quad Q^\top Q = I. \quad (16)$$

QR is chosen over SVD at insertion time because the operation requires only orthonormalization of the incoming batch — the number of columns is preserved, not reduced, so the ordered factorization that SVD provides is unnecessary overhead. SVD is reserved for the overflow handler: when the column count of G exceeds the budget M , the full basis is recompressed as $G \leftarrow U_{:,1:M}$, retaining the M left singular vectors of greatest variance and discarding the rest. This operation applies *at storage time* and affects only the basis G ; the gradient g entering the projection update itself is the unmodified parameter gradient from the model.

2. **Metric-tensor scaling.** Within each projection computation, all Fisher quantities are jointly divided by a single scale factor derived from the maximum entry of the method’s ambient metric F^* :

$$s = \max(F^*, 1), \quad \tilde{F} = F / s \text{ for } F \in \{F_{\text{old}}, F_{\text{new}}, F_c\}, \quad \tilde{\lambda} = \lambda / s, \quad (17)$$

where $F^* = F_{\text{new}}$ for FNG, ONG, and FOPNG, and $F^* = F_c$ for iFOPNG. The clamp to 1 prevents division below unity when the Fisher is near-zero. Critically, the damping coefficient λ is rescaled by the same factor, so its ratio to the Fisher magnitude is preserved across tasks regardless of how the Fisher scale drifts as more tasks accumulate. Because the projection kernel A , its pseudo-inverse, and the trust-region denominator are all computed under the same rescaled metric, the direction of v^* is invariant to s ; only the condition number of A is reduced, which is the sole purpose of the operation.

Together, these operations ensure that the method-specific projection kernel A remains well-conditioned regardless of the number of chunks spawned per task. The kernel differs structurally across methods:

$$A = \begin{cases} G^\top G + \lambda I & \text{OGD, ONG} \\ G^\top F_{\text{old}} F_{\text{new}}^{-1} F_{\text{old}} G + \lambda I & \text{FOPNG} \\ G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G + \lambda I & \text{iFOPNG} \end{cases} \quad (18)$$

Gradient orthonormalization eliminates column-scale inflation in G , which directly reduces the spectral range of A for all three variants. Metric-tensor normalization bounds $\tilde{F}$ to $[0, 1]$, preventing the Fisher-weighted variants from producing numerically singular A . The OGD/ONG kernel is unaffected by metric-tensor scaling by construction. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

4.6 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same inertial combined-metric preconditioner.

4.6.1 Benchmarks

Experiments are conducted on four standard continual learning benchmarks, following the selection of [Garg et al. \(2026\)](#) to enable direct numerical comparison.

Permuted-MNIST ([Kirkpatrick et al., 2017](#)) applies a distinct fixed random pixel permutation per task to a shared digit-classification head, evaluated as a 5-task stream in the standalone suite (Sub-RQ3) and as a 20-task long-horizon stream for the accumulation-operator comparison (Sub-RQ4).

Split-MNIST splits the 10-digit dataset into 5 binary tasks of 2 classes each; the multi-head protocol follows [Zenke, Poole, and Ganguli \(2017\)](#) and the single-head variant follows [Ven and Tolias \(2019\)](#), constituting the harder evaluation.

Split-CIFAR10 ([Zenke et al., 2017](#)) splits CIFAR-10 into 5 two-class tasks under a multi-head protocol, providing genuine visual inter-task interference.

Split-CIFAR100 (Zenke et al., 2017) extends this to 10 tasks of 10 classes each, testing scalability under a multi-head protocol. Hypernetwork configurations are evaluated on Split-MNIST, Split-CIFAR10, and Split-CIFAR100 with the architectural modifications described in Section 4.4.

4.6.2 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG, since the inertial combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (Yadav et al., 2025) is treated as a composite method: it applies OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries.

DEVIATION FOR FIRST-TASK INITIALIZATION. We deviated from the original protocol in one specific instance: first-task initialization for MNIST-based benchmarks. While Garg et al. (2026) state they used the method’s own learning rate for first-task SGD training (e.g., $\eta = 10^{-5}$ for FOPNG), our empirical replication of this protocol yielded an initial accuracy of only $\approx 21\%$. As demonstrated by the learning rate convergence sweep detailed in Appendix D (Figure 14), this published rate falls well below the empirical convergence threshold, effectively failing to establish a functional baseline. To ensure a clean initial projection subspace, we standardized the first-task SGD learning rate to 10^{-3} for MNIST-based benchmarks and 10^{-2} for CIFAR-based benchmarks. Beyond this necessary empirical correction, no tuning was performed on the published values.

HYPERNETWORK EXPERIMENTS. Settings not covered by Garg et al. (2026) follow the canonical hypernetwork framework of Oswald et al. (2020); exact values are detailed in Appendix 8. Because computing a full Fisher estimate requires a complete K -chunk generation pass per training sample, estimation is capped at 1,024 samples to maintain computational tractability — a standard approximation consistent with prior CIFAR-10 evaluations (Kirkpatrick et al., 2017). The gradient budget remains $K = 80$ per task, with memory compression handled via QR-and-SVD (Section 4.3). First-task training uses AdamW to seed the memory matrix G and Fisher estimate F from a clean geometric basis.

UNIFORM LEARNING RATE CONTROL. To isolate the projection constraint as the sole independent variable in the hypernetwork experiments, all methods use a uniform learning rate of 10^{-3} rather than method-specific rates. This prevents convergence-speed artifacts from conflating the structural effects of the projection geometry.

TRAINING DURATION AND SCHEDULING. Per-task training terminates at the first of: loss falling below 0.1; ten consecutive non-improving epochs; or the benchmark epoch cap (Appendix 8). A learning rate scheduler halves the magnitude after every three non-improving epochs; the best-loss parameter state is restored at termination. The rule is identical across all methods and consults only training loss, introducing no method-specific degree of freedom.

RANDOM SEEDS. All experiments draw from a fixed seed pool. Configurations evaluated on five seeds use $\{42, 111, 811, 1234, 2137\}$; the Split-CIFAR10 hypernetwork additionally includes seed 314, yielding $n = 6$ for that configuration; compute-limited configurations use the three-seed subset $\{42, 811, 1234\}$. The seed count for every reported comparison is stated alongside the corresponding result. One deviation applies: on the Split-MNIST hypernetwork benchmark, seed 2137 produced degenerate runs for every projection-based method while leaving the unconstrained baselines unaffected; it was therefore excluded and replaced by seed 314, yielding the five-seed set $\{42, 111, 314, 811, 1234\}$ used uniformly across all methods on this benchmark.

FISHER ACCUMULATION DEFAULT. The EMA decay parameter α appears exclusively in the Sub-RQ4 operator comparison ($\alpha = 0.5$); every other experiment accumulates F_{old} with the element-wise MAX operator, which introduces no additional hyperparameter.

STATISTICAL ANALYSIS. Pairwise comparisons (Sub-RQ3: iFOPNG vs. FOPNG; Sub-RQ4: MAX vs. EMA) use a two-tailed paired t -test over per-seed final average accuracy, with significance at $\alpha = .05$. Effect size is reported as Cohen’s $d_z = \bar{D}/s_D$, the mean per-seed difference divided by its sample standard deviation, so large values reflect seed-to-seed consistency rather than raw magnitude. The full eight-method comparison is reported descriptively rather than tested.

4.6.3 Sub-RQ1 — Constructive Synergy or Architectural Conflict

EXPERIMENTAL DESIGN. Sub-RQ1 investigates whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer

yields a net benefit, or if the loss of adaptive momentum makes a standard Adam-regularized hypernetwork superior. To separate architectural conflicts from benchmark-specific artifacts, we evaluate across three datasets at varying levels of compression, comparing an unconstrained standalone network (Panel a) against a hypernetwork (Panel b).

BENCHMARK SELECTION. We utilize a three-benchmark progression. **Benchmark 1 (Split-MNIST)** utilizes a deliberately “suffocated” hypernetwork (generator hidden dimension $d_h = 8$) to expose differential effects without hitting the near-perfect accuracy ceilings observed at higher capacities (see Appendix 8, Figure 11). To ensure convergence under these constraints, each task is trained for 15 epochs with a batch size of 64. **Benchmark 2 (Split-CIFAR10)** serves as the primary test, offering genuine inter-task interference and utilizing a standard-capacity hypernetwork trained for 50 epochs. **Benchmark 3 (Split-CIFAR100)** extends this sequence to 10 tasks to test scalability. Comparing performance within and across these panels reveals whether projection methods universally dominate, conditionally succeed based on compression, or systematically fail due to momentum stripping.

4.6.4 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is evaluated under three normalization conditions on the Split-CIFAR10 hypernetwork (Benchmark 2), with all other hyperparameters held constant. Split-CIFAR10 is chosen over the CIFAR-100 and MNIST hypernetwork configurations specifically because its chunk size of 6000 yields $C = 197$ chunks per target network — large enough to produce a measurable pathology, yet small enough that the condition number of A can be computed and logged at every projection step without becoming numerically intractable.

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With 197 chunks accumulated per forward pass, the gradient columns of G and the entries of F are expected to reach extreme magnitudes, inflating the condition

number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are rescaled by the maximum entry of the ambient metric F^* within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ of the projection kernel is logged before each inversion as the primary diagnostic. A large $\kappa(A)$ indicates that A is nearly singular: small perturbations in A are amplified by a factor of $\kappa(A)$ in the computed A^{-1} , causing the projected gradient to collapse to zero or NaN. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings, so any performance difference is attributable to normalization alone.

4.6.5 Sub-RQ3 — Does the Combined Fisher Metric Yield Consistent Retention Improvements

BENCHMARK SELECTION AND RATIONALE. Sub-RQ3 compares iFOPNG against FOPNG across the four standalone benchmarks evaluated by Garg et al. (2026): Permuted-MNIST (5 tasks), Split-MNIST single-head, Split-CIFAR10 multi-head, and Split-CIFAR100 multi-head. All hyperparameters are taken directly from Garg et al.’s Table 1, with the sole exception of the first-task learning rate for MNIST-based benchmarks, where the published value produces a degenerate initialisation; the necessary correction is established empirically in Appendix D (Figure 14). This replication-faithful design ensures that any observed difference between iFOPNG and FOPNG is attributable to the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ rather than to differences in training conditions.

Split-MNIST multi-head and the two hypernetwork configurations (Split-MNIST HN and Split-CIFAR10 HN) were already conducted for Sub-RQ1 and are included here for completeness; they are not part of

the Garg et al. replication and carry no additional inferential weight for Sub-RQ3.

4.6.6 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the inertial metric boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

For each sub-research question, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns. Average accuracy (A_T) is the primary evaluation metric, and backward transfer (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figures 7–9).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled. Plain SGD at 84.1% establishes the unconstrained reference, and Adam at 90.4% defines the upper frontier. iFOPNG leads all projection methods at 86.0%, the only one to surpass SGD. The remaining methods fall below the SGD reference: EWC at 83.1%, FNG at 81.1%, OGD at 79.6%, FOPNG at 77.9%, and ONG at 71.4%. Adam has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015).

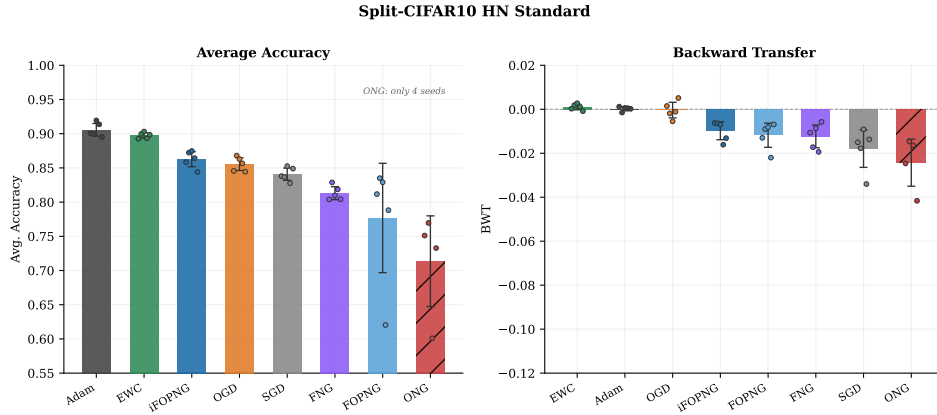


Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}).** Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained reference at 84.1%. iFOPNG is the only projection method to exceed SGD; the remaining projection variants fall below it. All experiments use full normalization (Section 5.2).

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. Figure 2 presents the full method comparison. Adam leads at 93.4%. The Fisher-free projection method OGD is the strongest projection baseline at 86.6%, followed by EWC at 84.5% and SGD at 79.8%. The Fisher-based methods fall sharply below: iFOPNG at 69.9%, FOPNG at 59.9%, and FNG at 54.8%. iFOPNG, FOPNG, and

FNG all achieve near-zero backward transfer (BWT ≈ 0.00), while Adam has small negative BWT (-0.015) and SGD forgets more substantially (-0.13).

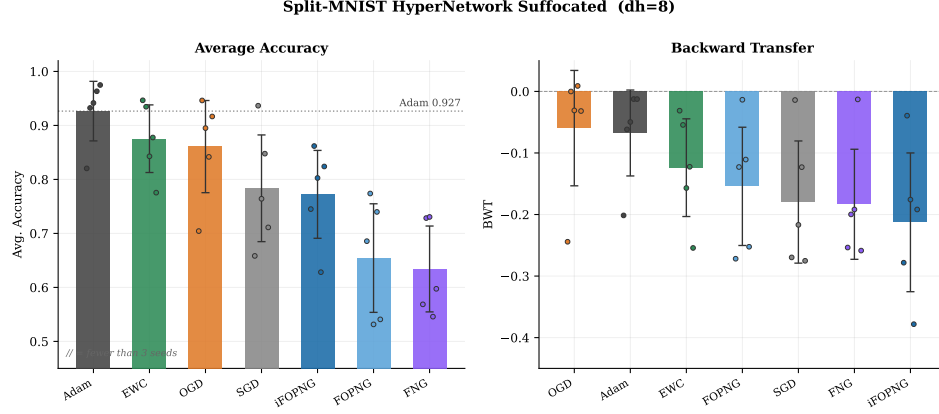


Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, AdamW first-task initialisation)**. Left: average accuracy (bars show mean over seeds; circles show individual seeds; note that several methods completed fewer than three seeds). Right: backward transfer. The dotted line marks Adam’s mean accuracy (93.4%). Fisher-based projection methods (iFOPNG, FOPNG, FNG) achieve near-zero BWT but much lower accuracy than Adam, revealing a plasticity failure rather than a retention failure in the extreme compression regime. The Fisher-free OGD is the best-performing projection method at 86.6%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 13), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%).

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation. Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top FG$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

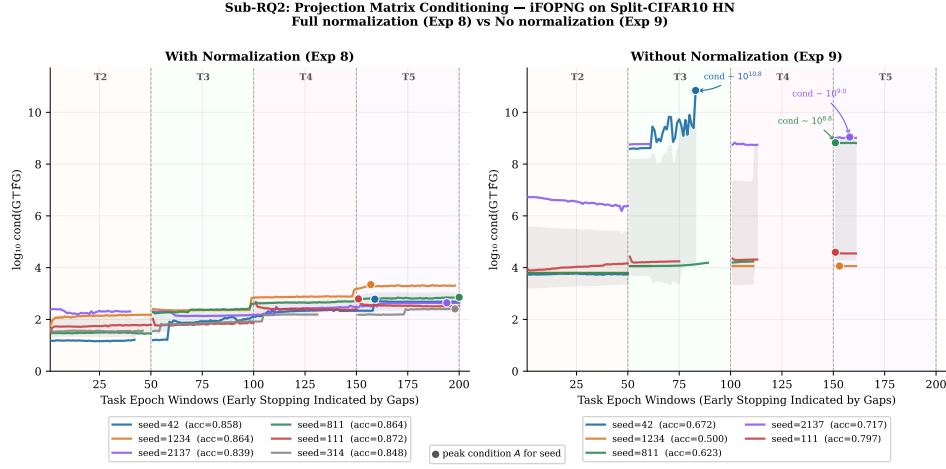


Figure 3: **Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(G^\top FG)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the $T_2 \rightarrow T_3$ transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

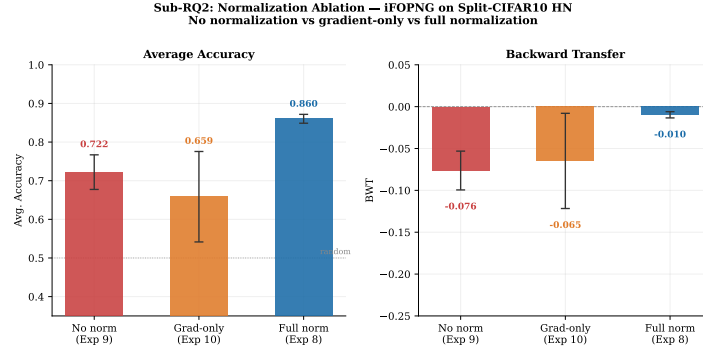


Figure 4: **Normalization ablation — iFOPNG on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the $T_2 \rightarrow T_3$ task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. The result is 72.2% average accuracy and $BWT = -0.070$.

GRADIENT-ONLY NORMALIZATION. Applying unit- ℓ_2 column normalization to G alone recovers 68.9% (BWT -0.060), below both full normalization (86.0%) and the no-normalization baseline (72.2%) (Figure 4).

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of F maintains $\log_{10} \kappa(G^\top FG)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions, a reduction of roughly seven orders of magnitude relative to the no-normalization condition (Figure 3). Average accuracy recovers to 86.0% and BWT improves to -0.013 (Figure 4).

5.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

Figure 5 presents mean average accuracy for iFOPNG and FOPNG across all eight benchmarks.

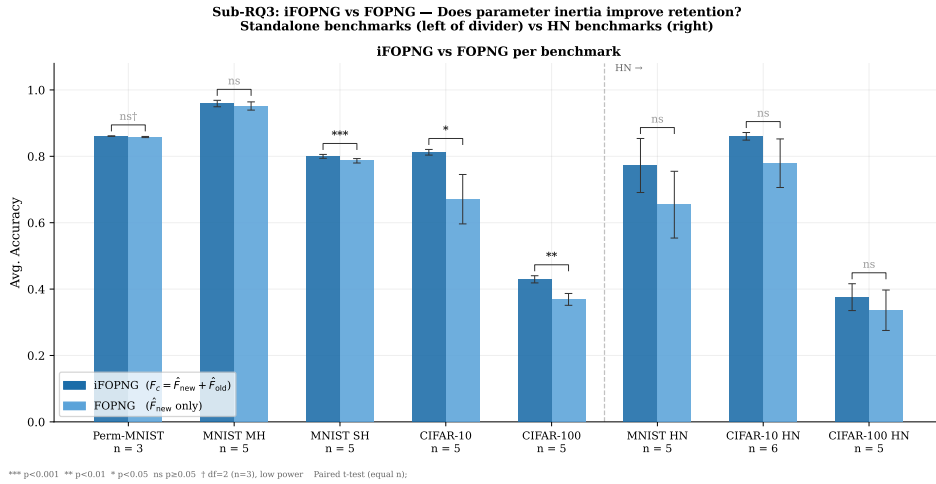


Figure 5: iFOPNG vs. FOPNG across eight benchmarks (Sub-RQ3). Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$; FOPNG uses only F_{new} .

On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (Permuted-MNIST: 86.1% vs. 85.8%, $t(2) = 3.89$, $p = .060$, $n = 3$; Split-MNIST MH: 95.9% vs. 95.2%, $t(4) = 1.12$, $p = .325$, $n = 5$). Using the task-incremental single-head protocol on Split-MNIST, iFOPNG reaches 80.0% versus FOPNG’s 78.7% ($t(4) = 24.07$, $p < 0.001$, $d_z = 10.76$). On Split-CIFAR10 multi-head, iFOPNG reaches 81.2% versus FOPNG’s 67.1% ($t(4) = 3.54$, $p = .024$,

$d_z = 1.58$, $n = 5$ seeds). On Split-CIFAR100 multi-head, iFOPNG reaches 42.9% vs. 36.9% ($t(4) = 6.55$, $p = .003$, $d_z = 2.93$, $n = 5$ seeds). In the three hypernetwork settings iFOPNG’s lead persists (Figure 5), but none of the three comparisons reaches statistical significance.

5.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer trajectories over all twenty tasks.

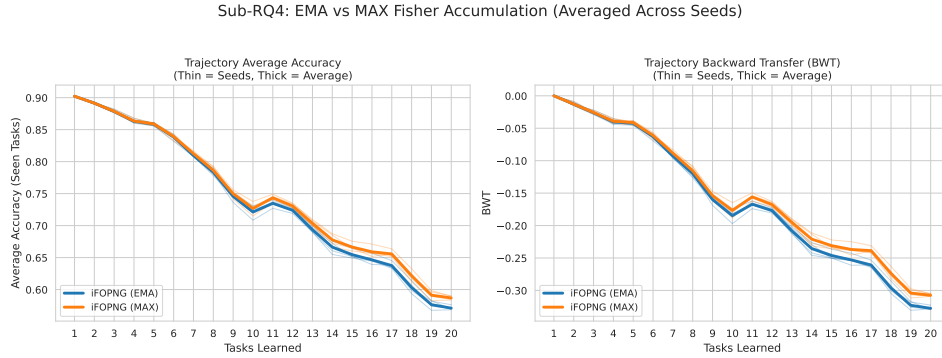


Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t -test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, $d_z = 6.45$. MAX leads EMA by 1.50–1.65 percentage points across every seed pair without exception.

6 DISCUSSION

The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via Adam. The gap of 4.4 pp at task 5 in favor of Adam is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp.

Critically, the BWT data reveal that in the suffocated regime this is a *plasticity* failure, not a *retention* failure: Fisher-based projection methods achieve near-zero backward transfer alongside collapsed accuracy, meaning the curvature-weighted constraint exhausts the available gradient budget for new task learning rather than permitting forgetting of prior ones. OGD’s relative resilience at $d_h = 8$ is consistent with this reading: without Fisher weighting its subspace constraint is less aggressive, leaving more gradient capacity for forward transfer. This extreme compression setting can be read as a proxy for the tight parameter budgets of edge-device deployment, suggesting that the plasticity collapse of Fisher-based methods under such budgets could limit their practical utility.

We attribute the broader gap to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer’s internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with [Hu, Yu, Cheng, Liu, and Song \(2026\)](#), who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: EWC unexpectedly leads at 45.5%, overtaking both Adam (42.0%) and iFOPNG (37.8%). As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. Standard Adam couples weight decay into the gradient as an additive ℓ_2 term, so both G and F collected at task 1 inherit a component aligned with weight magnitude rather than task loss curvature, contaminating the projection subspace. AdamW’s decoupled parameter-shrinkage step avoids this: it consistently produced lower first-task trajectory variance and a cleaner initial projection basis. For any system bootstrapping a projection memory from first-task gradients, the decoupled optimizer should be preferred at that stage.

6.2 Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology

The pathology is purely numerical rather than conceptual: the intersection of chunked hypernetworks and Fisher-weighted gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling. Although the ablation was conducted on iFOPNG, the conditioning problem is not specific to the combined metric F_c . The projection kernels of FOPNG and iFOPNG share the same algebraic structure — $A = G^\top F_{\text{old}} M^{-1} F_{\text{old}} G + \lambda I$, where $M = F_{\text{new}}$ for FOPNG and $M = F_c$ for iFOPNG — so the chunk-induced inflation of F_{old} degrades $\kappa(A)$ identically in both cases.

The normalization operates on the shared components G and F_{old} , and its effect on $\kappa(A)$ is therefore independent of which metric occupies the M^{-1} slot. OGD and ONG are structurally immune: their kernel $A = G^\top G + \lambda I$ contains no Fisher weighting, so gradient orthonormalization alone suffices and metric-tensor scaling is a no-op. This structural analysis implies that projection-scoped normalization is a necessary precondition for any method that embeds the Fisher metric inside the projection algebra — not an iFOPNG-specific patch.

Severity scales with chunk count C : gradient magnitudes accumulate as $C \cdot g_{\text{unit}}$, Fisher entries as $C^2 \cdot F_{\text{unit}}$, and the unnormalized kernel inflates by order C^4 per task. At sufficient scale the collapse is categorical rather than gradational — runs crash or converge to random-chance accuracy rather than merely underperforming — marking normalization as a structural stability prerequisite rather than a performance tuning. The elevated failure rate on Split-CIFAR100 hypernetwork, which has a larger chunk count than Split-CIFAR10, is consistent with this scaling argument even after full normalization is applied.

6.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

iFOPNG outperforms FOPNG in all eight settings, confirming that the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ yields a consistent *direction* of retention improvement. The gap is positive in every case, but its statistical strength tracks the degree of inter-task interference. It is unambiguous on the three standalone benchmarks that carry genuine interference: Split-MNIST single-head (+1.3 pp, $p < 0.001$, $d_z = 10.76$), Split-CIFAR₁₀ (+14.1 pp, $p = .024$), and Split-CIFAR₁₀₀ (+6.0 pp, $p = .003$).

It does not reach significance on the two near-ceiling benchmarks, where there is little to gain, nor in any hypernetwork setting, where inter-seed variance is high and the evidence is suggestive rather than conclusive. The magnitude varies across benchmarks but the direction does not, so Sub-RQ3 is answered affirmatively for the standalone setting and tentatively for the hypernetwork setting.

iFOPNG does not close the gap to Adam, confirming that the inertial combined metric improves retention by weighting historically important coordinates more heavily, but cannot recover the adaptive momentum that Adam leverages for efficient navigation of the generative manifold.

6.4 Sub-RQ4: Fisher Accumulation Operator

On a twenty-task Permuted-MNIST sequence, the element-wise maximum (MAX) accumulation strategy systematically outperformed the exponential moving average (EMA). While the 1.60 pp advantage is small, it is statistically unambiguous ($p < 0.001$).

Crucially, the hypothesized plasticity cost of MAX’s monotonic growth did not materialize. Because our normalization scheme rescales both Fisher terms, the projection becomes insensitive to absolute scale, relying solely on relative parameter distributions. Therefore, MAX remains the definitive practical choice: it is simpler, hyperparameter-free, and marginally superior, though highly heterogeneous task distributions could potentially introduce divergence.

6.5 Limitations

Several limitations should be acknowledged.

First, the hypernetwork results presented here are limited in task-sequence length: Split-MNIST and Split-CIFAR₁₀ use five tasks, and only the preliminary Split-CIFAR₁₀₀ result extends to ten. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization

may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the results are specific to the constrained-capacity regimes examined here. For Split-MNIST, a standard-capacity hypernetwork solves the benchmark near-trivially regardless of the continual learning method applied; the suffocated setting ($d_h = 8$) is deliberately chosen to place the architecture below saturation, where method differences are measurable. For Split-CIFAR10, the convolutional target network is large enough that the hypernetwork already operates under a natural capacity constraint, and meaningful method separation emerges without artificial bottlenecking. Whether the projection-hypernetwork interaction studied here generalises to harder benchmarks requiring larger target architectures — where the hypernetwork must generate weights for deeper convolutional or attention-based networks and the task itself is sufficiently complex to prevent ceiling effects — remains an open question. In such settings, the gradient accumulation pathology documented here would be substantially more severe: chunk counts scale with the target parameter count, so the condition number blowup $\sim C^4$ identified here represents a lower bound on the numerical challenge at scale.

Third, the proposed normalization scheme—QR orthonormalization combined with absolute-maximum Fisher scaling—is fundamentally heuristic in nature. While it successfully eliminates the numerical pathology of condition number collapse, it applies isotropic rescaling to the projection basis. This likely discards important geometric properties of the Fisher metric, temporarily divorcing the subspace constraint from the true curvature of the parameter space. A mathematically principled formulation that stabilizes the projection algebra while strictly preserving the theoretical guarantees of Fisher-based methods remains an open question for future work.

6.6 *Directions for Future Work*

The findings of this thesis highlight several avenues for future research. The most immediate extension is developing projection-compatible adaptive optimizers that maintain momentum across constrained updates (Hu et al., 2026).

Additionally, the heuristic QR normalization and FIM max scaling scheme must be replaced with a principled reformulation that integrates chunk-Jacobian curvature directly into the projection algebra.

Finally, future work should systematically evaluate these interactions on extended task sequences, where soft regularization penalties may degrade, and on larger target architectures, where the documented condition number blowup will require more robust mitigation.

7 CONCLUSION

This thesis investigated integrating gradient projection methods into chunked hypernetworks for continual learning. We primarily diagnosed the architectural boundaries and numerical pathologies governing this core integration. Alongside these fundamental structural findings, we also evaluated adjacent methodological optimizations regarding parameter inertia and Fisher matrix accumulation.

SUB-RQ1 — PARTIAL SYNERGY AND THE MOMENTUM BOTTLENECK. The synergy between hard gradient projection and soft functional regularization is constructive but bounded. While iFOPNG outperforms unconstrained SGD in standard-capacity settings, standalone regularized hypernetworks optimized via Adam remain vastly superior.

This performance gap stems from an architectural conflict, not projection geometry. Under extreme compression ($d_h = 8$), Fisher-based projection methods suffer a severe *plasticity* failure, exhausting the gradient budget and halting new task acquisition. Because projection modifies raw gradients before the optimizer updates, it discards adaptive momentum. In hypernetworks, this momentum acts as a crucial structural regularizer; without it, the network commits prematurely to poorly-conditioned gradient directions.

Ultimately, the loss of adaptive momentum is the binding constraint. Integrating gradient projection into hypernetworks will require projection-compatible adaptive optimizers, rather than post-hoc gradient modifications (Hu et al., 2026).

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods are numerically incompatible with chunked hypernetworks: matrix inversion in the projection kernel collapses completely, degrading performance below plain SGD. QR orthonormalization of the gradient basis combined with absolute-maximum Fisher scaling resolves the pathology entirely. This normalization scheme is a necessary engineering precondition for any future combination of gradi-

ent projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA VIA COMBINED FISHER METRIC. Replacing the task-specific Fisher metric with the combined inertial metric (F_c) improves retention whenever output-level interference is present. On the two near-ceiling benchmarks the two methods are statistically indistinguishable. On the standalone benchmarks that carry genuine interference, iFOPNG provides a clear, statistically significant advantage, largest on Split-CIFAR10 (+14.1 pp, $p = .024$). The same positive direction carries into the three hypernetwork settings, though none reaches significance under the elevated inter-seed variance of that regime. Integrating parameter inertia therefore yields a positive retention advantage in every setting: statistically robust on the standalone interference benchmarks, and directionally consistent but underpowered in the hypernetwork settings.

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. On a twenty-task sequence the element-wise maximum operator was statistically superior to an exponential moving average (~ 1.60 pp, $p < 0.001$), and the hypothesised plasticity cost of its monotonic growth did not materialise. MAX is therefore preferred: hyperparameter-free, simpler, and marginally better.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via the combined Fisher metric F_c , but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

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8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

Notation

Throughout this proof, $g \in \mathbb{R}^p$ denotes the current gradient, $G \in \mathbb{R}^{p \times m}$ the memory matrix of m historical gradient directions (each orthonormalised at insertion via QR; the proof does not rely on this property), and $F_{\text{new}}, F_{\text{old}}$ the current-task and accumulated diagonal empirical Fisher tensors, with $F_c = F_{\text{new}} + F_{\text{old}}$. Diagonal Fisher tensors are treated as vectors acting by element-wise multiplication and are identified with the corresponding diagonal matrices, which are symmetric. For a symmetric positive-definite S we write $\|w\|_S = \sqrt{w^\top S w}$. The projection kernel is

$$A := G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G \in \mathbb{R}^{m \times m}. \quad (19)$$

Assumption 1 (Regularity). F_c is positive definite and the kernel A is invertible.

Both conditions hold unconditionally in the implementation, which inverts the damped matrices $F_c + \hat{\lambda}I$ and $A + \hat{\lambda}I$ with $\hat{\lambda} > 0$ (Section 4.5.1); the theorem below states the exact $\lambda \rightarrow 0$ result.

Theorem 1 (iFOPNG Update). *Under Assumption 1, the solution to*

$$\begin{aligned} \min_u \quad & \|g - F_c^{-1}u\|_{F_c}^2 \\ \text{s.t.} \quad & F_{\text{old}}^{-1}u \in \text{Col}(G), \\ & \|v\|_{F_c} = \eta, \end{aligned} \quad (20)$$

with parameter update $v = c F_c^{-1}(g - u)$, is given by Equation (15):

$$v^* = -\eta \frac{F_c^{-1}Pg}{\sqrt{(Pg)^\top F_c^{-1}Pg}}, \quad P = I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}}. \quad (21)$$

Proof. **Step 1 (parameterise the constraint).** The constraint $F_{\text{old}}^{-1}u \in \text{Col}(G)$ means $F_{\text{old}}^{-1}u = G\alpha$ for some $\alpha \in \mathbb{R}^m$, i.e. $u = F_{\text{old}} G \alpha$. The optimisation over the constrained u thus reduces to an unconstrained optimisation over $\alpha \in \mathbb{R}^m$.

Step 2 (expand the objective). Writing $\|w\|_{F_c}^2 = w^\top F_c w$ and using $F_c^{-1}F_c = I$,

$$\|g - F_c^{-1}u\|_{F_c}^2 = g^\top F_c g - 2g^\top u + u^\top F_c^{-1}u. \quad (22)$$

Substituting $u = F_{\text{old}} G \alpha$, the linear term becomes $-2 (G^\top F_{\text{old}} g)^\top \alpha$ and the quadratic term becomes $\alpha^\top A \alpha$, since $G^\top F_{\text{old}} F_c^{-1} F_{\text{old}} G = A$. The first term $g^\top F_c g$ does not depend on α .

Step 3 (minimise). The reduced objective is quadratic in α with Hessian $2A \succ 0$, hence strictly convex. Setting its gradient to zero gives the normal equations $A \alpha^* = G^\top F_{\text{old}} g$, with unique solution

$$\alpha^* = A^{-1} G^\top F_{\text{old}} g. \quad (23)$$

Step 4 (back-substitute). Inserting α^* into $u = F_{\text{old}} G \alpha$ gives $u^* = F_{\text{old}} G A^{-1} G^\top F_{\text{old}} g$, so

$$g - u^* = (I - F_{\text{old}} G A^{-1} G^\top F_{\text{old}})g = Pg. \quad (24)$$

Step 5 (apply the trust region). For $v = c F_c^{-1} Pg$,

$$\|v\|_{F_c}^2 = c^2 (Pg)^\top F_c^{-1} F_c F_c^{-1} (Pg) = c^2 (Pg)^\top F_c^{-1} (Pg). \quad (25)$$

Imposing $\|v\|_{F_c} = \eta$ fixes $|c| = \eta / \sqrt{(Pg)^\top F_c^{-1} Pg}$. The negative root is selected so that with empty memory ($P = I$) the update reduces to the natural-gradient descent step under F_c , matching the FOPNG convention (Garg et al., 2026). This yields the stated update. $\square$

Remark (Relationship to FOPNG). The single substitution $F_{\text{new}} \rightarrow F_c$ recovers FOPNG: the proof is identical with F_c replaced by F_{new} in the metric, the kernel A , and the trust region. iFOPNG differs only in that the ambient metric carries the accumulated historical curvature F_{old} in addition to the current-task curvature.

Remark (Inertia mechanism). The trust-region budget splits additively across the two geometries:

$$\|v\|_{F_c}^2 = v^\top F_{\text{new}} v + v^\top F_{\text{old}} v = \|v\|_{F_{\text{new}}}^2 + \|v\|_{F_{\text{old}}}^2 = \eta^2. \quad (26)$$

A coordinate with large $(F_{\text{old}})_i$ consumes more of the shared budget η^2 per unit of motion and is therefore moved less. This is the sense in which historical importance acts as inertia rather than as a restoring force: no reference point and no penalty term enter the objective, in contrast to EWC (Kirkpatrick et al., 2017).

Algorithm Statements

Algorithm 2 is reproduced from Garg et al. (2026) without modification. It pre-multiplies each stored gradient by the Fisher estimate of the task it was collected on, absorbing the metric weighting into $\tilde{G}$ at collection

time so that $\tilde{G}$ approximates $F_{\text{old}} G$ (up to subsequent element-wise maximum updates of F_{old}) and the per-iteration projection reduces to a single matrix-vector product against $\tilde{G}$. Empirically the two algorithms produce indistinguishable results on all benchmarks evaluated here; preIFOPNG is the implementation preferred in practice due to its reduced cost.

Algorithm 1 iFOPNG Algorithm (extends FOPNG (Garg et al., 2026) via combined metric F_c)

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize gradient memory $G \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_{\text{old}} \leftarrow \hat{F}(D_1)$; Store $G \leftarrow [g_1, \dots, g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 under metric F_c
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Store gradients: $G \leftarrow [G, g_{k(t-1)+1}, \dots, g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional regularizer. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. Five benchmark streams are reported: Permuted-MNIST (5 tasks) and Split-MNIST multi-head (5 tasks) in Figure 7, Split-MNIST single-head (5 tasks) via per-task trajectory in Figure 10, Split-CIFAR10 multi-head (5 tasks) in Figure 8, and Split-CIFAR100 multi-head (10 tasks) in Figure 9.

Algorithm 2 preIFOPNG Algorithm (Garg et al., 2026), adapted for the iFOPNG combined metric

Require: Initial parameters θ_0 , task sequence $\{D_1, \dots, D_T\}$, learning rate η , damping λ , number of stored gradients per task k , number of epochs E

- 1: Initialize pre-Fisher memory $\tilde{G} \leftarrow []$. Train on task D_1 using Adam or SGD.
- 2: Compute $F_1 \leftarrow \hat{F}(D_1)$; $F_{\text{old}} \leftarrow F_1$; Store $\tilde{G} \leftarrow [F_1 \odot g_1, \dots, F_1 \odot g_k]$
- 3: **for** task $t = 2, 3, \dots, T$ **do**
- 4: **for** epoch $e = 1, \dots, E$ **do**
- 5: Recompute new-task Fisher: $\hat{F}_{\text{new}} \leftarrow \hat{F}(D_t)$;
- 6: Form combined metric: $F_c \leftarrow \hat{F}_{\text{new}} + F_{\text{old}}$
- 7: **for** all batches (x, y) in D_t **do**
- 8: Compute gradient $g \leftarrow \nabla_{\theta} \mathcal{L}_t(\theta)$
- 9: Compute v^* according to Theorem 1 using $\tilde{G}$ in place of G
- 10: Update parameters: $\theta \leftarrow \theta + v^*$
- 11: **end for**
- 12: **end for**
- 13: Compute $F_t \leftarrow \hat{F}(D_t)$; Store $\tilde{G} \leftarrow [\tilde{G}, F_t \odot g_{k(t-1)+1}, \dots, F_t \odot g_{kt}]$
- 14: Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}}, \hat{F}_{\text{new}})$
- 15: **end for**

Standalone Target-Network Benchmarks — MNIST

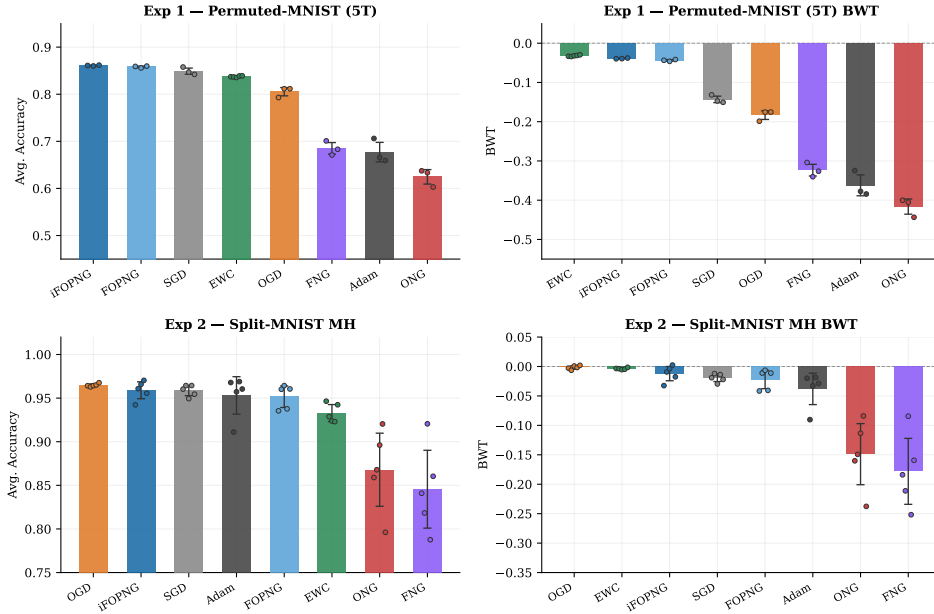


Figure 7: **Standalone target-network results — Permuted-MNIST (5 tasks, top row) and Split-MNIST multi-head (5 tasks, bottom row).** Left column: average accuracy. Right column: backward transfer. Bars show mean over seeds; circles show individual seeds. *Permuted-MNIST*. iFOPNG and FOPNG lead at 86.1% and 85.8% respectively with near-zero backward transfer, closely followed by SGD (84.9%) and EWC (83.7%). Adam (67.7%) and ONG (62.5%) exhibit substantially negative BWT (≈ -0.36 and ≈ -0.45), reflecting severe catastrophic forgetting without a projection or regularization mechanism. *Split-MNIST MH*. Most methods cluster near the ceiling ($\geq 95\%$), reflecting the low difficulty of the binary per-task splits. OGD, iFOPNG, SGD, and FOPNG lead (96.5–95.9%); ONG and FNG are the notable exceptions (86.8% and 84.6%) with strongly negative BWT, indicating convergence instability on this stream.

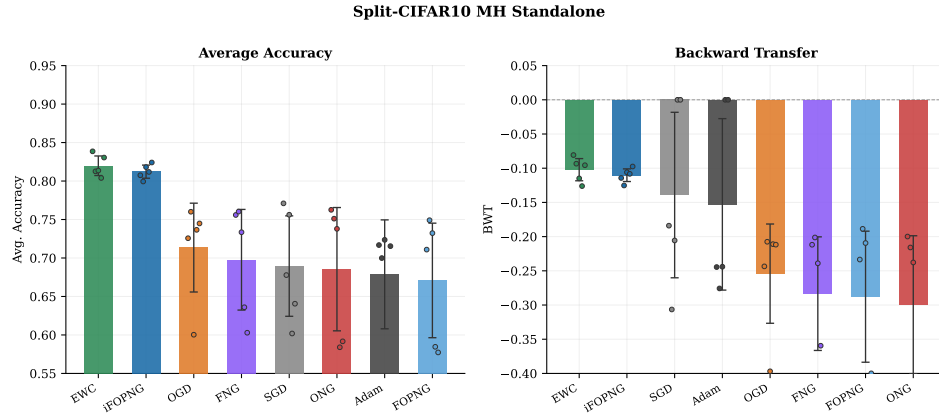


Figure 8: Standalone target-network results — Split-CIFAR₁₀ multi-head (5 tasks, 5 epochs per task, Adam first-task at 10^{-3}). Left: average accuracy. Right: backward transfer. EWC leads at 81.9% and iFOPNG is a close second at 81.2%; both exhibit moderate forgetting (BWT ≈ -0.09). The remaining methods fall to 67–71% with substantially higher seed variance and stronger forgetting. FOPNG is the weakest projection method at 67.4%, a 13.8 pp gap below iFOPNG that motivates the Sub-RQ₃ comparison. Adam (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks.

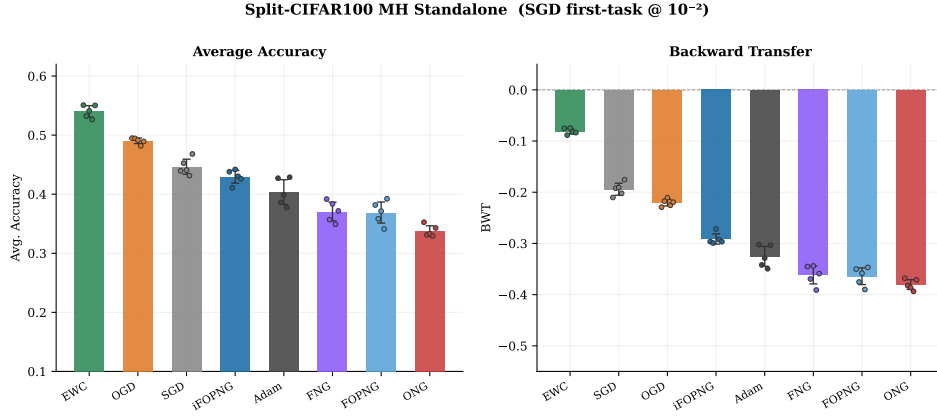


Figure 9: **Standalone target-network results — Split-CIFAR100 multi-head (10 tasks, 10 epochs per task, SGD first-task at 10^{-2})**. Left: average accuracy. Right: backward transfer. EWC ($\lambda=50$, full Fisher, $n = 5$ seeds; seed 111 not evaluated, seed 314 used instead) leads at 54.0%, followed by OGD (49.0%) and SGD (44.6%). iFOPNG reaches 42.9%, a 6.0 pp advantage over FOPNG (36.9%), consistent with the Sub-RQ3 finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. Adam (40.4%) sits slightly behind iFOPNG, while FNG (37.1%) closely mirrors FOPNG. ONG has the lowest average accuracy (33.8%) and the most negative BWT (≈ -0.38), indicating near-complete forgetting across the ten-task sequence. Random-chance performance is 10% (10-way classification per task).

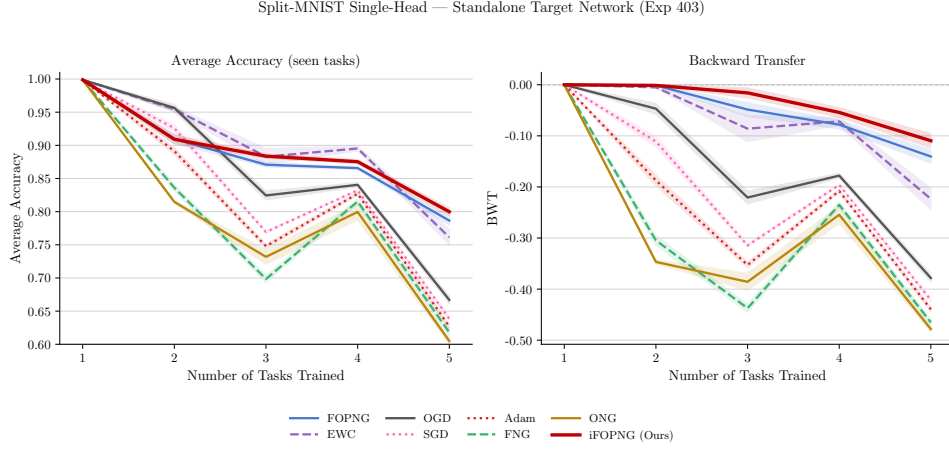


Figure 10: **Split-MNIST single-head standalone — all eight methods, 5 tasks, $n = 5$ seeds, task-incremental shared two-neuron output head (Sub-RQ₃).** Left: average accuracy over all tasks seen so far at each task step (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer from task 2 onward. iFOPNG maintains the highest average accuracy throughout all five task transitions, ending at 80.0% with BWT = -0.11 . FOPNG follows closely at 78.7% (BWT = -0.14), confirming the Sub-RQ₃ finding that the inertial combined-metric preconditioner provides a consistent advantage under output-level interference. EWC (76.1%) forms a second tier, with projection methods collectively outperforming all baselines. OGD (66.7%, BWT ≈ -0.38) performs substantially worse than on multi-head benchmarks, reflecting the harder gradient subspace overlap in the shared-head setting. The baselines SGD (63.8%), Adam (62.7%), FNG (61.8%), and ONG (60.5%) all show severe forgetting (BWT ≈ -0.42 to -0.48).

APPENDIX C: HYPERNETWORK ARCHITECTURE ABLATIONS

This appendix documents two architectural ablations that motivated the hypernetwork configurations used in the main experiments. Figure 11 establishes the diagnostic capacity level $d_h = 8$ for the suffocated Split-MNIST regime (Sub-RQ₁, Benchmark 1). Figure 12 presents the joint sweep over regularization strength β and chunk size c used to select the operating point for all main hypernetwork experiments.

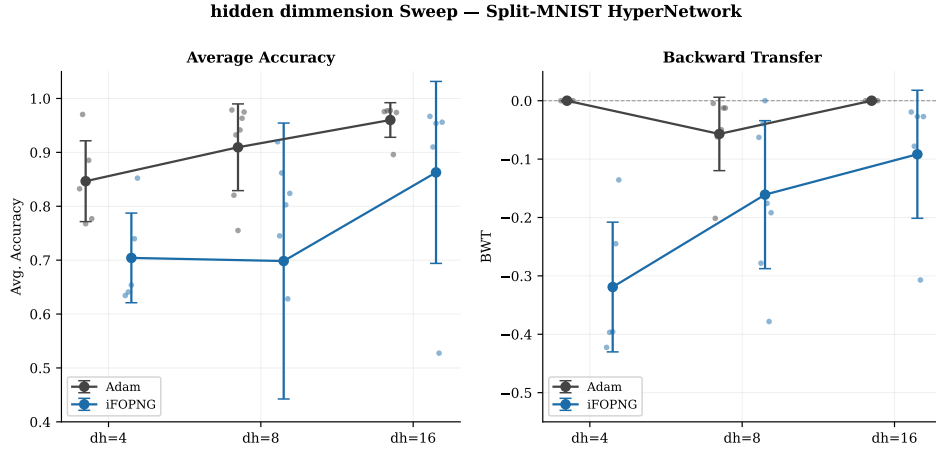


Figure 11: **Hidden-dimension sweep — Split-MNIST SH hypernetwork** ($d_h \in \{4, 8, 16\}$, 5 tasks, 15 epochs per task, AdamW first-task initialisation at 10^{-3}). Left: average accuracy; right: backward transfer. Thick markers show seed means; error bars show ± 1 standard deviation; individual seeds are shown as dots. Only Adam and iFOPNG are plotted to isolate the projection–baseline gap across capacity levels. Three capacity levels are compared. At $d_h = 4$, the Adam–iFOPNG gap is 14.0 pp, but individual seed variance for iFOPNG is extreme (range ≈ 0.25), indicating that the bottleneck is too severe for reliable projection convergence. At $d_h = 8$, the gap widens to 23.5 pp and variance contracts substantially: seeds cluster in the range 0.63–0.92 for Adam and 0.63–0.75 for iFOPNG, making the failure mode severe and consistently replicable. At $d_h = 16$, projection methods partially recover — the gap narrows to 9.4 pp — as the larger gradient budget reduces the relative cost of subspace projection. The backward transfer panel mirrors this pattern: iFOPNG BWT improves monotonically with capacity (≈ -0.32 at $d_h = 4$, ≈ -0.17 at $d_h = 8$, ≈ -0.11 at $d_h = 16$), while Adam remains near zero throughout. $d_h = 8$ was retained as the primary diagnostic level: the plasticity failure is both severe and consistently replicable, whereas $d_h = 4$ yields near-random results in some seeds and $d_h = 16$ partially obscures the failure mode.

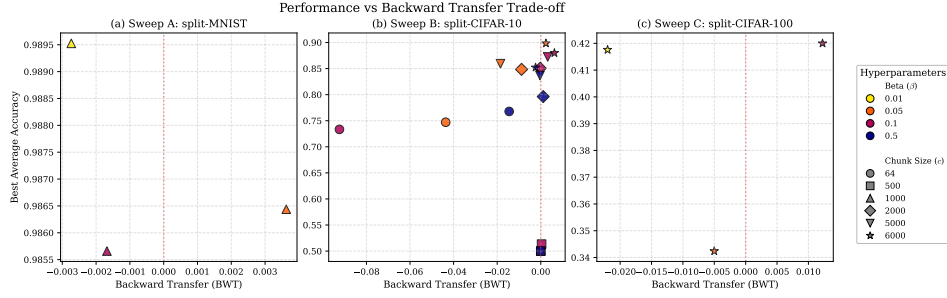


Figure 12: Beta and chunk-size sweep — hypernetwork performance vs. backward transfer trade-off across three benchmarks. Each panel plots mean average accuracy (y -axis) against backward transfer (x -axis) for all evaluated combinations of regularization strength $\beta \in \{0.01, 0.05, 0.1, 0.5\}$ and chunk size $c \in \{64, 500, 1000, 2000, 5000, 6000\}$. Marker shape encodes chunk size; marker colour encodes β . The vertical red dashed line marks $\text{BWT} = 0$; points to its left exhibit net retention, points to its right exhibit net forgetting. The probe method is Adam with the functional regularizer active. Split-MNIST (panel a) is insensitive to both hyperparameters: all combinations achieve 98.55–98.95% accuracy with BWT spanning only ≈ 0.006 , indicating that the task is too simple to discriminate architectures. Split-CIFAR₁₀ (panel b) shows clear structure. Small chunk sizes ($c \in \{64, 500\}$) cluster in the lower-left regardless of β , reaching only 74–75% accuracy; larger chunks progressively shift the Pareto frontier upward. The combination $c = 6000$, $\beta = 0.1$ sits on the efficiency frontier: it achieves 87–88% average accuracy with small negative BWT, providing the best retention–plasticity balance. The combination $c = 6000$, $\beta = 0.5$ yields similar accuracy but with marginally higher forgetting. Split-CIFAR₁₀₀ (panel c) confirms the selection: $c = 6000$, $\beta = 0.5$ achieves the highest accuracy ($\approx 42\%$) but at the cost of slightly positive BWT, suggesting the stronger regularizer over-constrains plasticity on a 10-task sequence. $c = 6000$, $\beta = 0.1$ achieves $\approx 34\%$ with negative BWT and was therefore selected as the operating point for all main hypernetwork experiments, providing consistent Pareto-frontier performance across both CIFAR benchmarks without the positive-BWT artefact observed at higher β .

APPENDIX D: ADDITIONAL EXPERIMENT FIGURES

This appendix collects seven figures that are referenced in the Results section but omitted from the main text for space. Figure ?? provides the per-task accuracy matrix for the Split-CIFAR₁₀ hypernetwork experiment. Figure 13 shows the full method comparison on the ten-task Split-CIFAR₁₀₀ hypernetwork benchmark. Figure 14 documents the SGD first-task convergence diagnostic. Figure 15 reports the 0GD learning-rate sensitivity sweep. Figure 16 provides the per-seed paired comparison underlying the Sub-RQ₄ statistical test. Figure 17 compares the SVD, FIFO, and STOP gradient-memory compression strategies on 20-task Permuted-MNIST. Figure 18 documents the seed-2137 initialization failure on the Split-MNIST

hypernetwork benchmark that motivated the seed-replacement policy of Section 4.6.2.

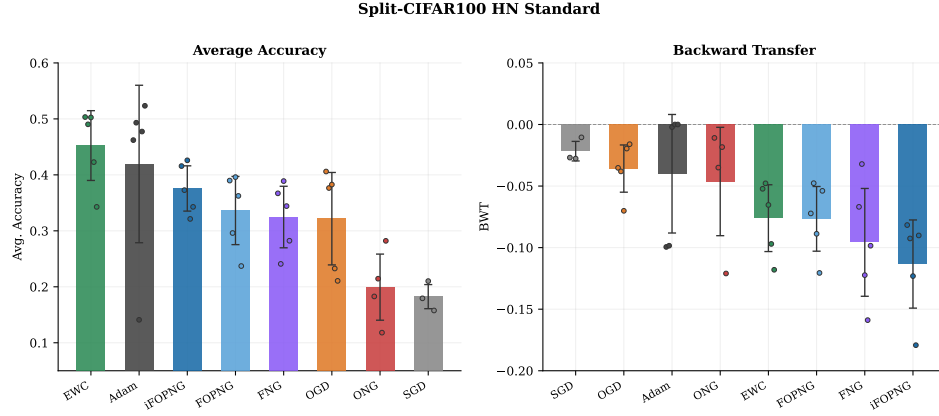


Figure 13: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3} .** Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). EWC leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform EWC and closely trail Adam). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, Adam contains one outlier seed at $\approx 14\%$, pulling the mean below the median. SGD (18.7%) and ONG (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

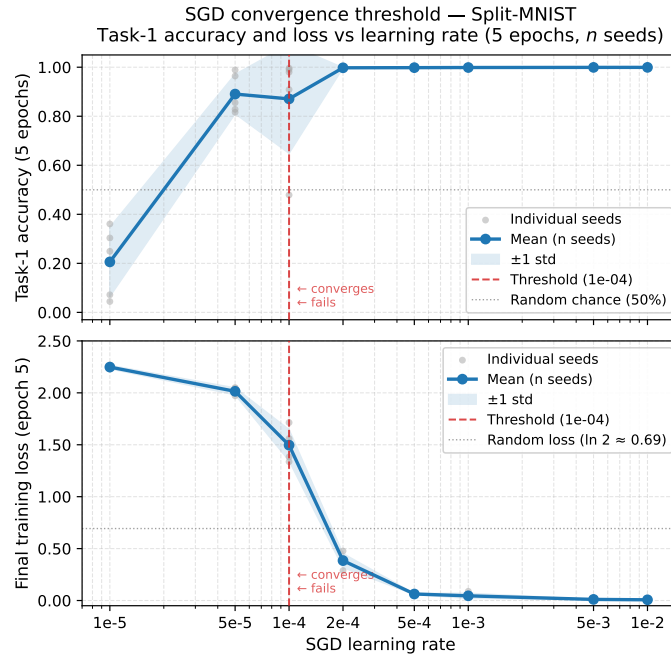


Figure 14: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only)**. Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

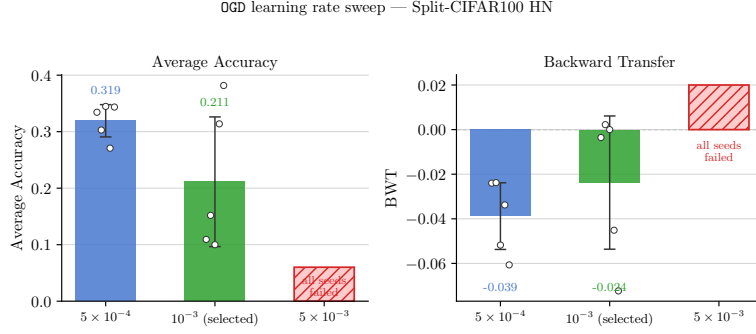


Figure 15: **OGD learning rate sweep — Split-CIFAR100 hypernetwork (10 tasks, standard capacity), $n = 3$ seeds per rate.** Left: final average accuracy. Right: final backward transfer. Bars show seed mean; error bars show ± 1 standard deviation; circles show individual seeds. $\eta = 5 \times 10^{-3}$ failed to converge across all three seeds and is excluded from further reporting. $\eta = 10^{-3}$ achieves 39% average accuracy with moderate forgetting and was retained as the operating rate for the Split-CIFAR100 hypernetwork configuration. $\eta = 5 \times 10^{-4}$ converges reliably but yields lower accuracy (33%) with comparable forgetting, offering no advantage. This sweep is the sole instance in which a method-specific learning rate required empirical selection beyond the values specified in Garg et al. (2026) Table 1.

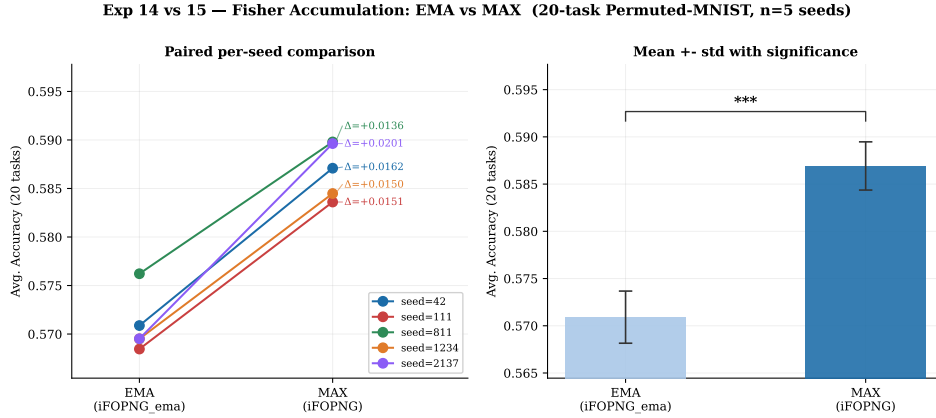


Figure 16: **EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4).** Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t -test significance bracket. Every seed shows a positive slope (MAX > EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from +1.50 to +1.65 pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen's $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

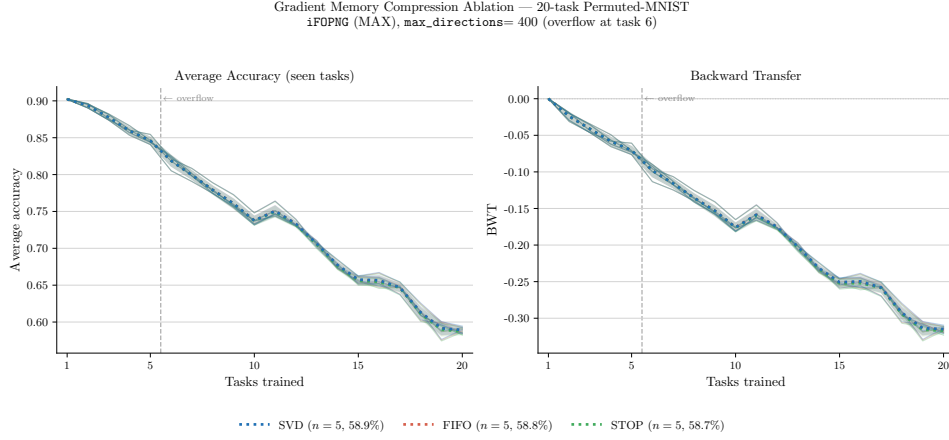


Figure 17: **Gradient memory compression ablation — 20-task Permuted-MNIST, iFOPNG with MAX accumulation, max_directions=400, $n=5$ seeds each.** Left: average accuracy trajectory over 20 tasks (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer trajectory from task 2 onward. The vertical dashed line marks the first compression event at task 6, at which point the gradient basis is full and each new task’s directions displace existing ones according to strategy. Three strategies are compared: SVD (retain top- K singular vectors), FIFO (evict oldest task directions), and STOP (freeze basis, discard new directions). All three trajectories are visually indistinguishable throughout; final average accuracy spans 58.71%–58.93%, a between-condition range of 0.22 pp. The result confirms that allocation strategy is dominated by budget size: with 400 directions for 20 tasks (≈ 20 per task), the binding constraint is the budget-to-task ratio rather than how the budget is distributed. On Permuted-MNIST, where task distributions are structurally homogeneous, all gradient directions carry equal importance and SVD’s importance-weighted selection provides no measurable advantage over the simpler alternatives.

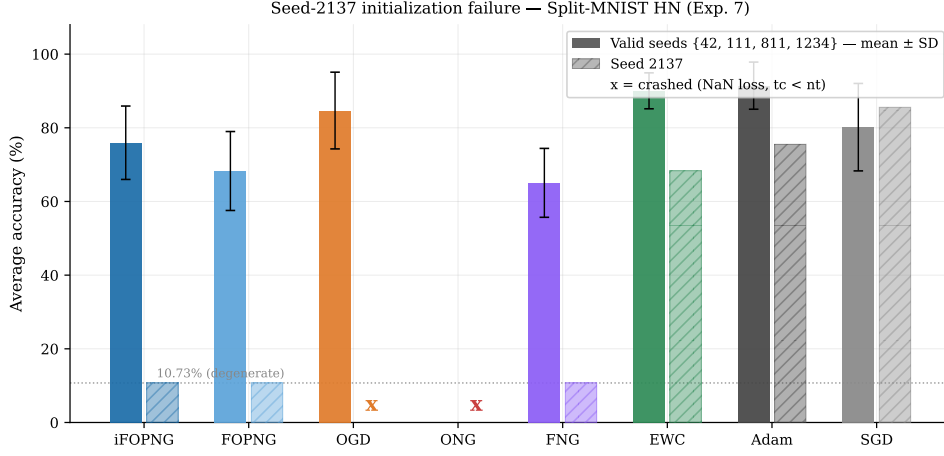


Figure 18: **Seed-2137 initialization failure — Split-MNIST hypernetwork ($d_h = 8, 5$ tasks, task-incremental single-head).** Each group of bars shows average accuracy for one method; within each group the lighter bar is the mean over the four valid seeds $\{42, 111, 811, 1234\}$ (± 1 SD, shown as error bar) and the darker bar is the value obtained under seed 2137. Methods whose seed-2137 bar is absent (ODG, ONG) crashed with NaN train loss at task 1 and produced no evaluable result. The failure is selective by method class. Unconstrained baselines (Adam, SGD) and EWC show no anomaly under seed 2137, converging normally and posting results within the valid-seed distribution. Every gradient-memory method (FNG, FOPNG, iFOPNG) converges to an identical degenerate accuracy of 10.73% — a value equal to the single-task mean across the five tasks, indicating that task 1 was learned normally (53.7% of final accuracy attributable to task 1 alone) while every subsequent task was never learned. The underlying mechanism is a shared task-1 initialization failure. Under this seed, all adaptive-optimizer runs of the hypernetwork converge to a degenerate task-1 solution at 53.7% accuracy (barely above binary chance on the 5-task single-head protocol) rather than the $\approx 99\%$ reached by SGD on the same seed. Methods that carry nothing forward from task 1 (Adam, EWC) recover by learning tasks 2–5 independently. Methods that bootstrap their gradient memory G and Fisher estimate $\hat{F}$ from the task-1 solution are permanently foreclosed: the projection subspace is seeded from a near-random initialisation, causing all subsequent gradient updates to be projected to near-zero. OGD and ONG additionally overflow during Fisher estimation on the untrained task-2 embedding, producing NaN propagation consistent with the diagnostic in the thesis body. Because the failure is caused by the initialization and affects all projection-based methods identically, it renders any cross-method comparison on this seed invalid; seed 2137 was therefore replaced by seed 314 for the Split-MNIST hypernetwork experiment (see Section 4.6.2).

APPENDIX F: EXTENDED RESULTS

This appendix provides a complete numerical reference for all eight methods across all eight benchmarks evaluated in this thesis. Table 1 reports

seed-averaged final average accuracy; formal statistical comparisons are reserved for the targeted iFOPNG vs. FOPNG analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

Table 1: **Mean average accuracy (%) across all benchmarks and methods.** Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use target networks trained directly; HN benchmarks use the chunked hypernetwork architecture. [†]MNIST HN is the suffocated regime ($d_h = 8$). [‡]CIFAR-100 MH EWC used $\lambda = 10$; $\lambda = 50$ collapsed and is excluded. *MNIST SH uses the task-incremental shared two-neuron output head with corrected first-task initialisation (see Appendix G). **Bold**: best per column. ***Bold italic***: iFOPNG (ours). —: insufficient seeds or unavailable.

Method	Standalone					Hypernetwork		
	Perm-MNIST	MNIST MH	MNIST SH*	CIFAR-10 MH	CIFAR-100 MH [‡]	MNIST HN ($d_h=8$) [†]	CIFAR-10 HN	CIFAR-100 HN
<i>iFOPNG</i> (ours)	86.1	95.9	80.0	81.2	42.9	69.9	86.0	37.8
FOPNG	85.8	95.2	78.7	67.1	36.9	59.9	77.9	33.8
EWC	83.7	93.3	76.1	82.0	46.6	84.5	83.1	45.5
OGD	80.5	96.5	66.7	71.3	49.0	86.6	79.6	32.3
FNG	68.5	84.6	61.8	69.8	37.5	54.8	81.1	32.5
ONG	62.5	86.8	60.5	68.5	33.8	—	71.4	20.1
SGD	84.9	95.9	63.8	69.0	45.1	79.8	84.1	18.7
Adam	67.7	95.3	62.7	67.9	40.4	93.4	90.4	40.1
Random chance	50.0	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX G: HYPERPARAMETER TABLES

All values are sourced directly from the W&B run configs (CSV logs) unless marked [?] (inferred from thesis text or figure annotations). Entries marked [†] are flagged in Section ??; `max_ep` is the maximum epochs per task; `ep/task` is the configured training epochs; `max_dirs` is the gradient memory column budget.

G.1 Standalone Target-Network Experiments

All standalone experiments share the following fixed settings: `model = TargetNetwork`, `normalize = False`, `regulizer = False`, `damping = 0.01`, `batch_size = 10`, $n = 5$ seeds. Learning rates are taken from Garg et al. (2026) Table 1 and vary per method; they are reproduced in Table 3 for reference. The first-task optimizer is SGD for all MNIST benchmarks and Adam for CIFAR benchmarks, at $\eta_{\text{first}} = 10^{-3}$ throughout. For MNIST benchmarks this deviates from the published value of $\eta_{\text{first}} = 10^{-5}$, which fails to converge (see Appendix D, Figure 14).

Table 2: **Benchmark-level training constants.** Fisher samples marked *full* use the entire training set (60 000 for Permuted-MNIST; 12 000 for Split-MNIST).

Benchmark	Tasks	ep/task	Fisher samp.	EWC λ	Proj. λ
Permuted-MNIST	5	5	<i>full</i> (60 000)	10	10^{-2}
Split-MNIST MH	5	5	<i>full</i> (12 000)	400	5×10^{-4}
Split-MNIST SH	5	5	<i>full</i> (12 000)	400	5×10^{-4}
Split-CIFAR10 MH	5	5	1 024	50	10^{-3}
Split-CIFAR100 MH	10	10	1 024	$10^\dagger$	10^{-3}

† CIFAR-100 MH: $\lambda = 50$ produced collapsed runs and is excluded; $\lambda = 10$ is used (see Figure 9).

Table 3: **Per-method learning rates** taken from Garg et al. (2026) Table 1. iFOPNG uses the same values as FOPNG.

Benchmark	Adam	SGD	EWC	FNG	OGD	ONG	FOPNG	iFOPNG
Permuted-MNIST	10^{-4}	5×10^{-3}	10^{-4}	10^{-3}	5×10^{-3}	5×10^{-3}	10^{-4}	10^{-4}
Split-MNIST MH/SH	10^{-5}	5×10^{-4}	5×10^{-4}	10^{-3}	5×10^{-4}	5×10^{-4}	10^{-5}	10^{-5}
Split-CIFAR10 MH	10^{-3}	5×10^{-2}	10^{-3}	10^{-2}	5×10^{-2}	10^{-2}	10^{-3}	10^{-3}
Split-CIFAR100 MH	10^{-4}	5×10^{-3}	10^{-2}	5×10^{-3}	10^{-2}	10^{-2}	5×10^{-3}	5×10^{-3}

G.2 Hypernetwork Experiments

All HN experiments: model = HyperNetwork, regularizer = True, $\beta = 0.1$ ($\text{lam} = 0.001$), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 4: **Hypernetwork architecture and training parameters.** c_emb = chunk embedding dim; t_emb = task embedding dim; max_dir = maximum gradient memory directions. † Exp 411: results extracted from the 35 completed runs at chunk = 6 000, $d_h = 32$, $normalize = \text{True}$; remaining runs are exploratory pilots (see G.6).

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task	max_dir
CIFAR-10 HN (main, Exp 408)	5	10	50	AdamW	32	6 000	16	16	80	1 000
CIFAR-100 HN (Exp 411) †	10	10	50	AdamW	32	6 000	16	16	200	800
MNIST HN ($d_h=8$, suffocated, Exp 407)	5	15	15	AdamW	8	64	32	32	100	5 000
d_h sweep — $d_h = 4$	5	15	15	AdamW	4	64	32	32	80	5 000
d_h sweep — $d_h = 8$	5	15	15	AdamW	8	64	32	32	100	5 000
d_h sweep — $d_h = 16$	5	15	15	AdamW	16	64	32	32	80	5 000

G.3 Normalization Ablation

Table 5: **Normalization ablation conditions (Sub-RQ2)**. All conditions: Split-CIFAR10 HN, chunk = 6 000, $d_h = 32$, iFOPNG only, AdamW first-task at $\eta = 10^{-3}$. Exp 410 (gradient-only) contains pilot runs at chunk = 256, $d_h = 64$; only chunk = 6 000, $d_h = 32$ runs are used.

Condition	Exp	normalize	QR on G	Max-scale $\hat{F}$	Seeds	Accuracy	BWT
No normalization	409	False	×	×	5	72.2%	−0.042
Gradient-only	410	True	✓	×	3	68.9%	−0.060
Full normalization	408	True	✓	✓	6	86.0%	−0.006

G.4 Sub-RQ4 and Compression Ablation

Both Sub-RQ4 conditions and all three compression ablation conditions use 20-task Permuted-MNIST, TargetNetwork, normalize = False, batch = 10.

Table 6: **Sub-RQ4 accumulation parameters** ($n = 5$ seeds, seeds {42, 111, 811, 1234, 2137}, $\alpha = 0.5$ for EMA).

Operator	Tasks	ep/task	max_ep	LR	λ	grads / task	Fisher samp.
MAX (iFOPNG)	20	5	5	10^{-4}	10^{-2}	80	60 000
EMA (iFOPNG_ema, $\alpha = 0.5$)	20	5	5	10^{-4}	10^{-2}	80	60 000

Table 7: **Compression ablation parameters** ($n = 5$ seeds each, seeds {42, 111, 811, 1234, 2137}). $\lambda = 10^{-3}$ differs from the Sub-RQ4 baseline ($\lambda = 10^{-2}$); the ablation is internally controlled but not directly comparable to Sub-RQ4 on this hyperparameter. $\alpha = 0.3$ for the internal F_old EMA update in all three conditions, differing from the Sub-RQ4 value of $\alpha = 0.5$; results are therefore not directly comparable to Sub-RQ4 on this parameter either.

Strategy	Tasks	ep/task	max_ep	LR	λ	grads / task	max_dirs
SVD (Exp 421)	20	5	5	10^{-4}	10^{-3}	80	400
FIFO (Exp 422)	20	5	5	10^{-4}	10^{-3}	80	400
STOP (Exp 423)	20	5	5	10^{-4}	10^{-3}	80	400